

A0. OOBN for asset failure due to internal corrosion of oil and gas pipeline				
Parameters		Sub criteria	State	Discretization Values
A1. A Bayesian subnetwork model for UDC (Fig.4)				
1	UDC rate (mm/year)	Operating parameter; Fluid chemistry; Material parameter; Deposition parameter	High	$2 \leq \text{UDC rate} < 5$
			Moderate	$1 \leq \text{UDC rate} < 2$
			Low	$\text{UDC rate} < 1$
2	Operating parameter	Temperature; pH; Operating pressure; Wetting factor	Absent	CPT
			Present	
3	Temperature (T), °C		High	$60^{\circ}\text{C} \leq T$
			Moderate	$30^{\circ}\text{C} \leq T < 60^{\circ}\text{C}$
			Low	$T < 30^{\circ}\text{C}$
4	pH		Acidic	$6 \leq \text{pH}$
			Neutral	$5 \leq \text{pH} < 6$
			Alkaline	$4 \leq \text{pH}$
5	Operating pressure (OP), Bar		High	$100 \leq \text{OP}$
			Moderate	$10 \leq \text{OP} < 100$
			Low	$\text{OP} < 10$
6	Wetting factor	Water cut; Flow velocity; Flow type	False	CPT
			True	
7	Water cut (WC), %		High	$30 \leq \text{WC} < 100$
			Moderate	$15 \leq \text{WC} < 30$

			Low	WC < 15
8	Flow type (FT), m/s	Density; Pipe internal diameter; Viscosity	High	$2 \leq FT < 3$
			Moderate	$1 \leq FT < 2$
			Low	$0.1 \leq FT < 0.5$
			Stagnant	$0 \leq FT < 0.1$
9	Density		High	Case dependent
			Moderate	
			Low	
10	Pipe internal diameter, (mm)		More than 481.5mm	Database
			Diameter 1 to 481.4mm	
11	Viscosity, cp		High	Case dependent
			Moderate	
			Low	
12	Fluid chemistry	Redox potential; Chemical parameter	Absent	CPT
			Present	
13	Redox potential	Electron acceptor presence; Electron donor presence; O ₂ ; Hydrogen	Low	CPT
			High	
14	Electron acceptor presence		False	Case dependent
			True	
15	Electron donor presence		False	Case dependent
			True	

16	O ₂ , ppm		High	$100 \leq O_2$
			Moderate	$10 \leq O_2 < 100$
			Low	$O_2 < 10$
17	Hydrogen		High	Unknown
			Moderate	
			Low	
18	Iron oxide	Fe ₂ plus; Fe ₃ plus; O ₂	Absent	CPT
			Present	
19	Fe ₂ plus (Fe ²⁺), ppm		Non-reactive	$0 \leq Fe^{2+} < 30$
			Reactive	$30 \leq Fe^{2+} < 100$
20	Fe ₃ plus (Fe ³⁺), ppm		Non-reactive	$0 \leq Fe^{3+} < 40$
			Reactive	$40 \leq Fe^{3+} < 100$
21	Corrosion parameter	Iron oxide; Carbonates; Sulphates	Absent	CPT
			Present	
22	Chemical parameter	Chloride; H ₂ S; Corrosion parameter; CO ₂	Low	CPT
			High	
23	Chloride (Cl), ppm		High	$30,000 \leq Cl$
			Moderate	$500 \leq Cl < 30,000$
			Low	$0 \leq Cl < 500$
24	H ₂ S, ppm		Low	$0 \leq H_2S < 50$
			High	$50 \leq H_2S < 1000$

25	CO ₂ , Bar		High	$10 \leq \text{CO}_2 < 100$
			Moderate	$1 \leq \text{CO}_2 < 10$
			Low	$0 \leq \text{CO}_2 < 1$
26	Material parameter	Surface parameter; Design parameter	No	CPT
			Yes	
27	Surface parameter	Surface; Surface coating; Welding marks	No	CPT
			Yes	
28	Surface		Absent	Pipeline operator
			Present	
29	Surface coating		No	Pipeline operator
			Yes	
30	Welding marks		No	Pipeline operator
			Yes	
31	Design parameter	Bends and inclination; Discontinuities and dead	No	CPT
			Yes	
32	Bends and inclination		Inclined	Pipeline operator
			Flat	
33	Discontinuities and dead		No	Pipeline operator
			Yes	
34	Organic deposits	Wax; Asphaltenes	No	CPT
			Yes	
35	Wax		No	Database
			Yes	
36	Asphaltenes		No	Database

			Yes	
37	Inert deposits	Scale presence; Sand presence; Debris presence	No	CPT
			Yes	
38	Scale presence		High	Depends on the product type
			Moderate	
			Low	
39	Sand presence		False	Asset specific parameters
			True	
49	Debris presence		No	Asset specific parameters
			Yes	
41	Deposit presence	Inert deposits (as above); Organic deposits (as above); Metallic oxides; Biofilm presence; Deposition parameter	No	CPT
			Yes	
42	Deposition parameter	Deposit elimination; Deposit depth; Deposit porosity; Deposit presence (as above)	High	CPT
			Moderate	
			Low	
43	Deposit porosity		No	Case dependent
			Yes	
44	Deposit elimination	Inspection state; Pigging effectiveness; Pigging frequency	False	CPT
			True	
45			No	Pipeline operator

	Inspection state		Yes	
46	Pigging effectiveness		Low	Pipeline operator
			High	
47	Pigging frequency		High	- More than Once 4 weeks
			Moderate	- Once 12 weeks
			Low	- Less than once 48 weeks
48	Metallic oxides		No	Unknown
			Yes	
49	Biofilm presence		No	Unknown
			Yes	
A2. A Bayesian subnetwork for uniform corrosion (Fig.5)				
50	Uniform corrosion	Inhibited corrosion rate; Wetting factor	False	CPT
			True	
51	Inhibited corrosion rate	Uninhibited corrosion rate; Corrosion inhibition	CPT	CPT
52	Corrosion inhibition		No	Unknown
			Yes	
53	Internal corrosion mitigation	Corrosion inhibition; Internal coating	False	CPT
			True	
54	Uninhibited corrosion rate	Iron oxide; H ₂ S; CO ₂ ; Temperature; pH	No	CPT
			Yes	

A3. Bayesian network for localized corrosion (Fig.6)				
55	Localized corrosion	Galvanic cell; UDC rate	False	CPT
			True	
56	UDC rate	Deposit presence; Wetting factor	High	CPT
			Medium	
			Low	
57	Galvanic cell	Anodic film; Passive film; Chloride; Localized inhibitor	No	CPT
			Yes	
58	Localized inhibitor	Corrosion inhibition; Flow velocity	False	CPT
			True	
59	Anodic film		False	Unknown
			True	
60	Passive film		False	Unknown
			True	
A4. A Bayesian subnetwork model for erosion (Fig.7)				
61	Erosion corosion	Solid particles; Water cut; Geometry change	No	CPT
			Yes	
62	Solid particles		High	Depends on the product type
			Moderate	
			Low	
63	Geometry change	Valve;	No	CPT

		Bends & inclination; T_piece; Diameter change	Yes	
64	Diameter change		No	Unknown
			Yes	
65	T_piece		No	Pipeline operator
			Yes	
66	Valve		No	Pipeline operator
			Yes	
A5. A Bayesian subnetwork model for MIC (Fig.8)				
67	MIC (mm/year)	Operating parameter; Bacteria Presence; Wetting factor; Water condition; MIC control	No	CPT
			Yes	
68	Operating parameter	Deposit presence; Operating parameter; Pigging frequency	False	CPT
			True	
69	Bacteria Presence		False	Unknown
			True	
70	Water condition, ppm	Redox potential; Mineral content	False	CPT
			True	
71	MIC control	Pigging frequency; Biocides treatment; Pigging effectiveness	False	CPT
			True	
A6. A Bayesian subnetwork model for asset failure of a gas pipeline (Fig.9)				
72	Asset failure	Operating pressure;	No	CPT

		Pipe outer diameter; Pipe wall thickness	Yes	
73	Pipe outer diameter, mm		OD 497.4 to 512.4	Database
			512.5 to 514.4 mm	
74	Pipe wall thickness	Asset failure due to; UDC_Rate (UDC rate) Asset failure due to uniform corrosion (Uniform corrosion); Asset failure due to localized corrosion (Localized corrosion); Asset failure due to erosion (Erosion); Asset failure due to MIC (MIC).	False	CPT
			True	
CPT: Based on conditional probability tables developed by experts' knowledge elicitation (Step 4).				

Case history parameters in DOOBN model

Important Parameters	Value
Temperature (T), °C	28-45°C
pH	Up to 5.31
Operating pressure (OP), Bar	120 - 140 Bar
Water cut (WC), %	Moderate
Flow type (FT), m/s	Gas
Density	Up to 1025 kg/m ³
Pipe internal diameter, (mm)	479.4 mm
Viscosity, cp	1.8- 5 cp
Fluid chemistry	Experts' knowledge
Electron acceptor presence	N/A
Electron donor presence	N/A

O ₂ , ppm	N/A
Hydrogen	N/A
Fe ₂ plus (Fe ²⁺), ppm	N/A
Fe ₃ plus (Fe ³⁺), ppm	N/A
Chloride (Cl), ppm	Up to 6540 ppm
H ₂ S, ppm	Up to 20
CO ₂ , Bar	Up to 100
Surface	N/A
Surface coating	N/A
Welding marks	N/A
Bends and inclination	Flat
Discontinuities and dead	Yes
Wax	High
Asphaltenes	High
Scale presence	Moderate
Sand presence	Less sand
Debris presence	N/A
Deposit porosity	N/A
Inspection state	Yes
Pigging effectiveness	Good
Pigging frequency	Intelligent pig Less than once 48 weeks
Metallic oxides	N/A
Biofilm presence	N/A

Corrosion inhibition	N/A
Anodic film	N/A
Passive film	N/A
Solid particles	High
Diameter change	N/A
T_piece	Yes
Valve	Yes
Bacteria Presence	High
Pipe outer diameter, mm	514.4 mm